NUMERICAL COMPUTING II

Assignment II

Which Numerical Differentiation Method Should We Trust?

1. Aim

The aim of this assignment is to investigate the accuracy of three numerical differentiation methods: Forward Difference, Backward Difference, and Central Difference. Their performance is compared using three functions and different step sizes to determine which method provides the most accurate results and how the step size affects numerical error.

2. Theory

Numerical differentiation estimates the derivative of a function using function values at nearby points. It is useful when an exact derivative is difficult to obtain or when only numerical function values are available.

Forward Difference:

$f'(x) \approx [f(x + h) - f(x)] / h$

Assignment2 > forward_difference.py > ...

```python

class ForwardDifference:
    def derivative(self, f, x, h):
        return (f(x + h) - f(x)) / h

```

Backward Difference:

$f'(x) \approx [f(x) - f(x - h)] / h$

Assignment2 > backward_difference.py > ...

```python

class BackwardDifference:
    def derivative(self, f, x, h):
        return (f(x) - f(x - h)) / h

```

Central Difference:

$f'(x) \approx [f(x + h) - f(x - h)] / (2h)$

```
Assignment2 > central_difference.py > ...
1
2   class CentralDifference:
3       def derivative(self, f, x, h):
4           return (f(x + h) - f(x - h)) / (2 * h)
5
```

Here, x is the evaluation point and h is the step size. The absolute error is calculated as |exact derivative − numerical derivative|. Forward and Backward Difference generally have first-order truncation error, O(h), while Central Difference generally has second-order truncation error, $O(h^2)$.

3. Experimental Setup

The methods were implemented in Python using three separate classes. The math module was used to evaluate the functions, and Matplotlib was used to plot absolute error against step size on logarithmic axes. The evaluation point was x = 1.

Parameter	Value
Function 1	$f(x) = e^x$
Function 2	$f(x) = \sin(x)$
Function 3	$f(x) = x^2 - 2x + 1$
Step sizes	10^{-1}, 10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}, 10^{-6}
Evaluation point	x = 1
Error measure	Absolute error

4. Exact Derivatives

Function	Exact derivative	Value at x = 1
e^x	e^x	2.718281828459
sin(x)	cos(x)	0.540302305868
$x^2 - 2x + 1$	$2x - 2$	0

4.1 Function 1: $f(x) = e^x$

Step size (h)	Forward Difference	Backward Difference	Central Difference
10^{-1}	1.4056×10^{-1}	1.3149×10^{-1}	4.5327×10^{-3}
10^{-2}	1.3637×10^{-2}	1.3546×10^{-2}	4.5305×10^{-5}
10^{-3}	1.3596×10^{-3}	1.3587×10^{-3}	4.5305×10^{-7}
10^{-4}	1.3592×10^{-4}	1.3591×10^{-4}	4.5306×10^{-9}
10^{-5}	1.3591×10^{-5}	1.3591×10^{-5}	5.8587×10^{-11}
10^{-6}	1.3590×10^{-6}	1.3593×10^{-6}	1.6346×10^{-10}

Observation: Forward and Backward Difference have similar errors that decrease approximately in proportion to h. Central Difference produces much smaller errors. Its error reaches about 5.86×10^{-11} at $h = 10^{-5}$, then increases at $h = 10^{-6}$, indicating the effect of floating-point round-off.

4.2 Function 2: f(x) = sin(x)

Step size (h)	Forward Difference	Backward Difference	Central Difference
10^{-1}	4.2939×10^{-2}	4.1138×10^{-2}	9.0005×10^{-4}
10^{-2}	4.2163×10^{-3}	4.1983×10^{-3}	9.0050×10^{-6}
10^{-3}	4.2083×10^{-4}	4.2065×10^{-4}	9.0050×10^{-8}
10^{-4}	4.2074×10^{-5}	4.2073×10^{-5}	9.0043×10^{-10}
10^{-5}	4.2074×10^{-6}	4.2073×10^{-6}	1.1141×10^{-11}
10^{-6}	4.2075×10^{-7}	4.2080×10^{-7}	2.7717×10^{-11}

Observation: Central Difference is more accurate for every tested step size. Its minimum observed error is approximately 1.11×10^{-11} at $h = 10^{-5}$. At $h = 10^{-6}$, the error increases because round-off error becomes more significant.

4.3 Function 3: f(x) = $x^2 - 2x + 1$

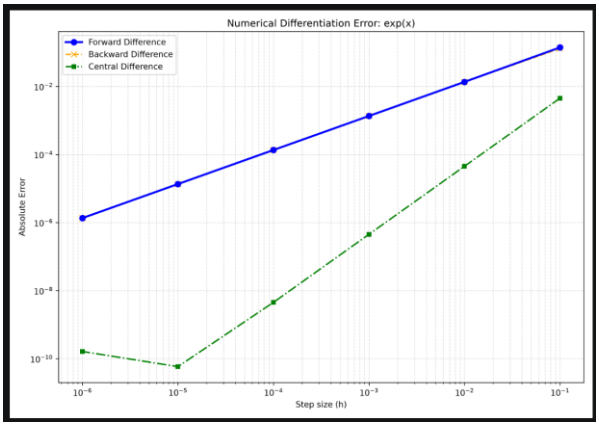
Step size (h)	Forward Difference	Backward Difference	Central Difference
10^{-1}	1.0000×10^{-1}	1.0000×10^{-1}	0
10^{-2}	1.0000×10^{-2}	1.0000×10^{-2}	0
10^{-3}	1.0000×10^{-3}	1.0000×10^{-3}	5.5511×10^{-14}
10^{-4}	1.0000×10^{-4}	1.0000×10^{-4}	5.5511×10^{-13}
10^{-5}	1.0000×10^{-5}	1.0000×10^{-5}	0
10^{-6}	1.0001×10^{-6}	9.9998×10^{-7}	5.5511×10^{-11}

Observation: Forward and Backward Difference produce errors approximately equal to h. Central Difference is exact for a quadratic function in exact arithmetic, because the central-difference formula cancels the higher-order terms. Tiny nonzero errors in the output are caused by floating-point arithmetic. Zero-error points cannot be shown on a logarithmic error axis.

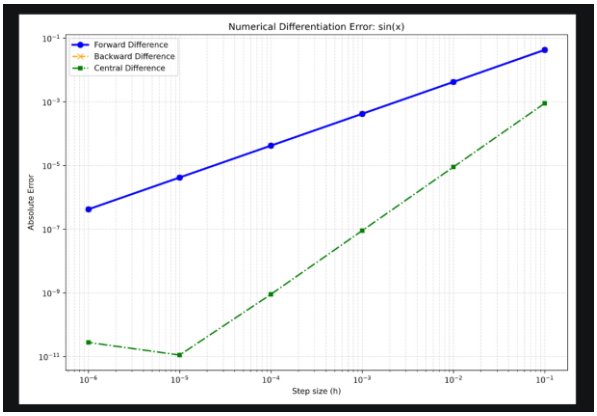
5. Analysis of the Log-Log Plots

The log-log plots compare step size with absolute error. Forward and Backward Difference generally have curves with a slope close to 1 in the truncation-error-dominated region, while Central Difference has a slope close to 2. This supports the expected first-order and second-order convergence behaviour. The Forward and Backward curves may overlap visually because their errors are very similar. Different line styles and markers can make both methods easier to distinguish.

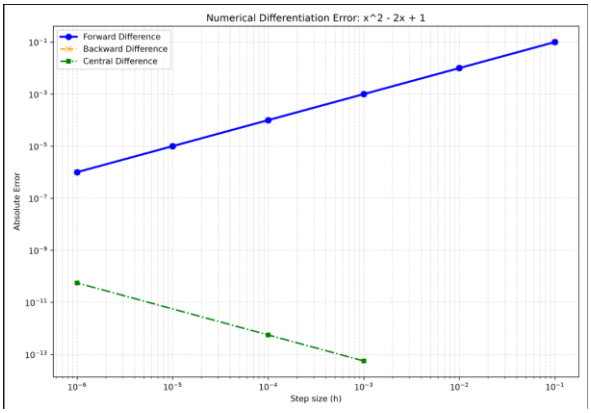
Insert the three plots generated by the program here:



- Figure 1. Numerical differentiation error for $f(x) = e^x$.



- Figure 2. Numerical differentiation error for f(x) = sin(x).



- Figure 3. Numerical differentiation error for f(x) = $x^2 - 2x + 1$.

6. Answers to the Assignment Questions

Q1. Which method is consistently the most accurate?

Central Difference is the most accurate method in these experiments. For e^x and sin(x), it gives much smaller absolute errors than Forward and Backward Difference at every tested step size. For the quadratic function, it gives zero or near-zero error. This is due to its second-order truncation error, compared with the first-order truncation errors of Forward and Backward Difference. Its accuracy still depends on choosing a suitable step size.

Q2. Does the observed error decrease at the expected rate?

Yes, the results generally follow the theoretical rates. Forward and Backward Difference errors decrease approximately in proportion to h, while Central Difference errors decrease approximately in proportion to h^2 when truncation error dominates. For example, for e^x, reducing h from 10^{-1} to 10^{-2} reduces the Central Difference error from about 4.53×10^{-3} to 4.53×10^{-5}, a reduction by roughly a factor of 100.

Q3. Is there a value of h beyond which the error stops decreasing?

Yes. For e^x and sin(x), the Central Difference error is smallest among the tested values at $h = 10^{-5}$, then increases at $h = 10^{-6}$. At very small step sizes, subtraction of nearly equal floating-point numbers can amplify round-off error. The best step size depends on the function, method, and machine precision.

Conclusion

The experiment shows that numerical differentiation accuracy depends on both the method and the selected step size. Forward and Backward Difference give similar results, with errors generally decreasing linearly as h decreases. Central Difference produces smaller errors because it has a higher order of accuracy. For the quadratic function, it is exact in mathematical arithmetic, apart from negligible floating-point effects in the computed results. The experiment also shows that reducing h indefinitely is not always beneficial, because round-off error can eventually increase the total error. Therefore, Central Difference is the most reliable of the three methods for these tests when an appropriate step size is selected.

Submitted By,

Khumba Lunganlung,MS2515